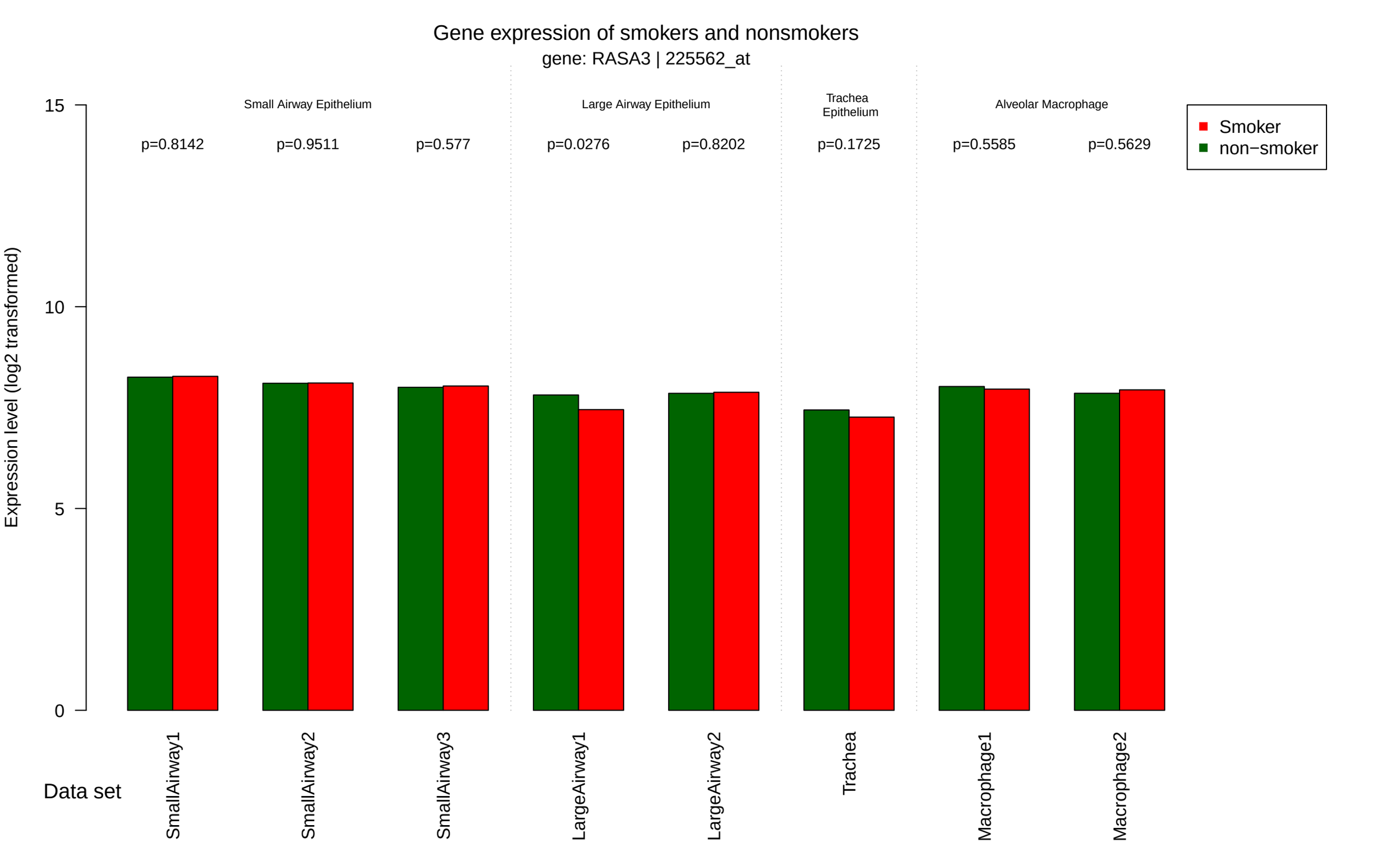
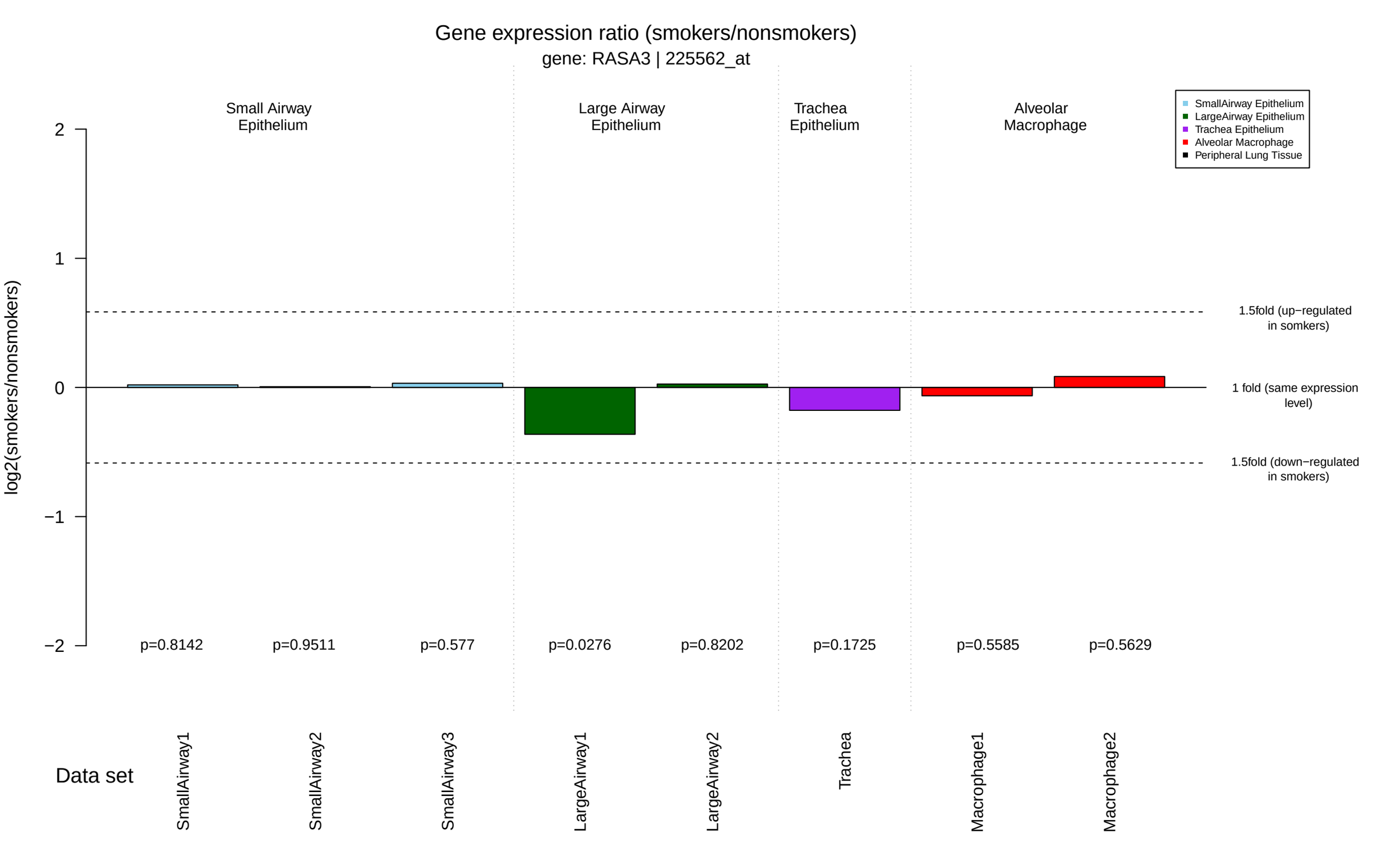
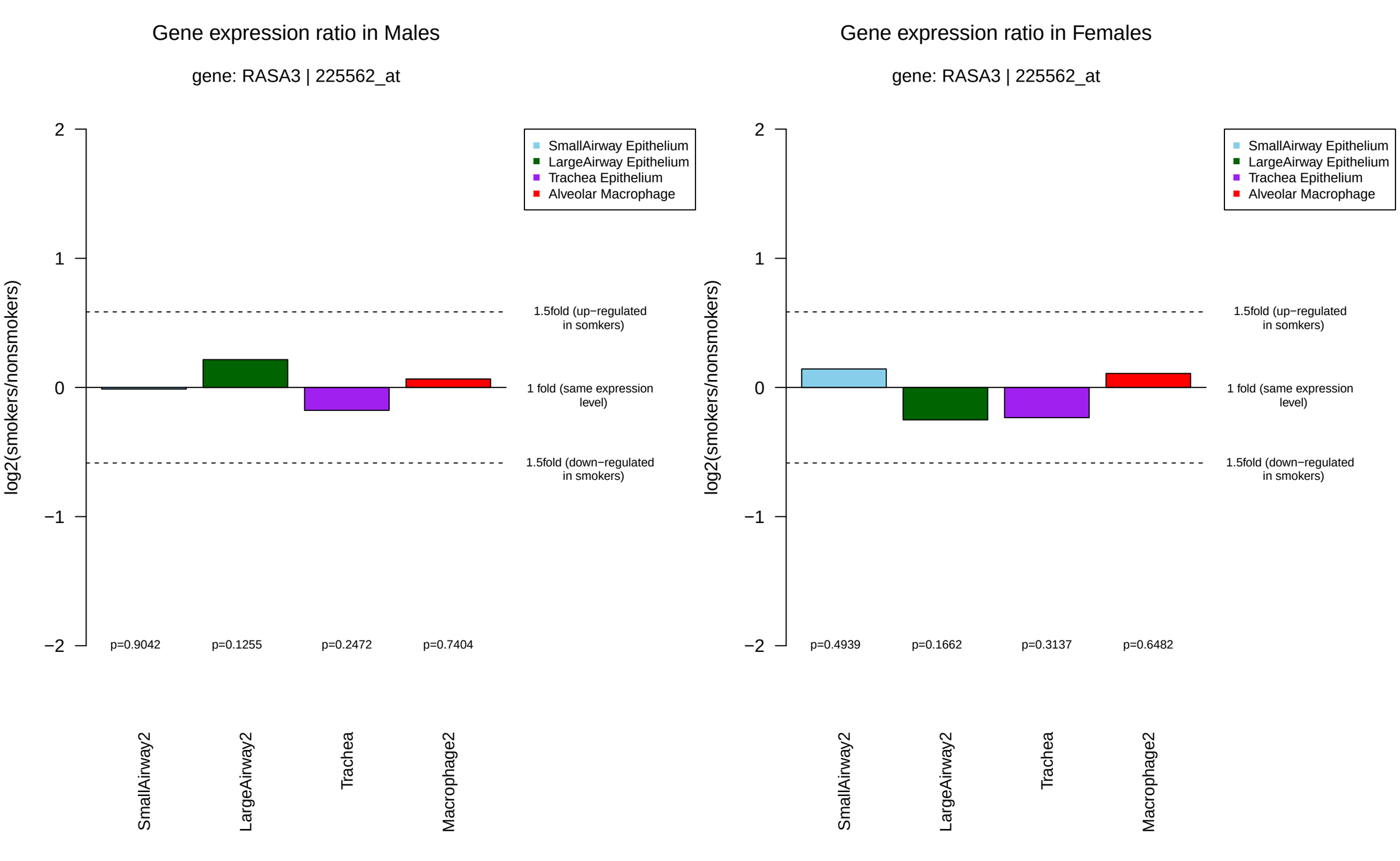
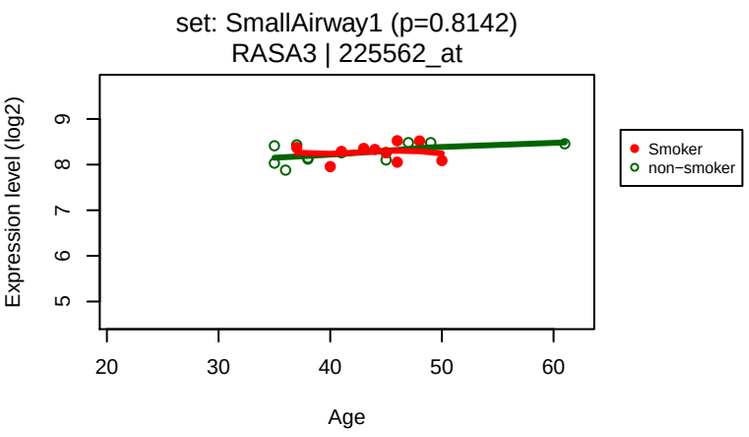
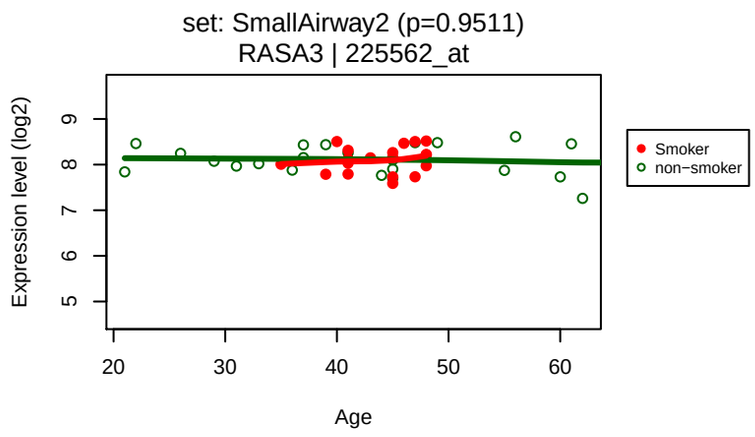
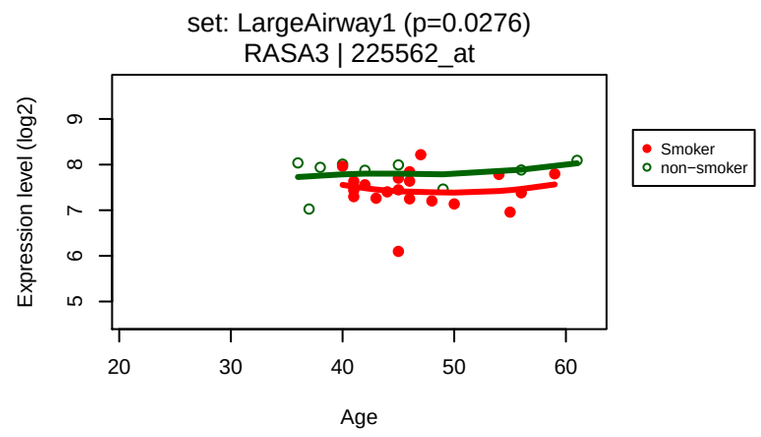

Gene expression of smokers and nonsmokers
gene: RASA3 | 225562_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8142
p=0.9511
p=0.577
p=0.0276
p=0.8202
p=0.1725
p=0.5585
p=0.5629
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

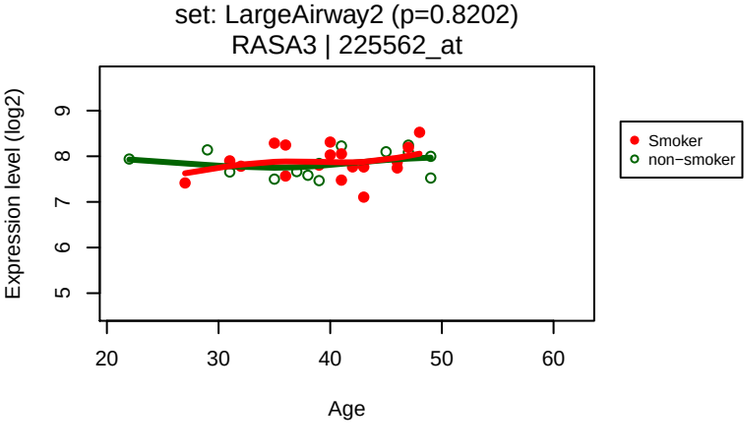

Gene expression ratio (smokers/nonsmokers)
gene: RASA3 | 225562_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8142
p=0.9511
p=0.577
p=0.0276
p=0.8202
p=0.1725
p=0.5585
p=0.5629
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

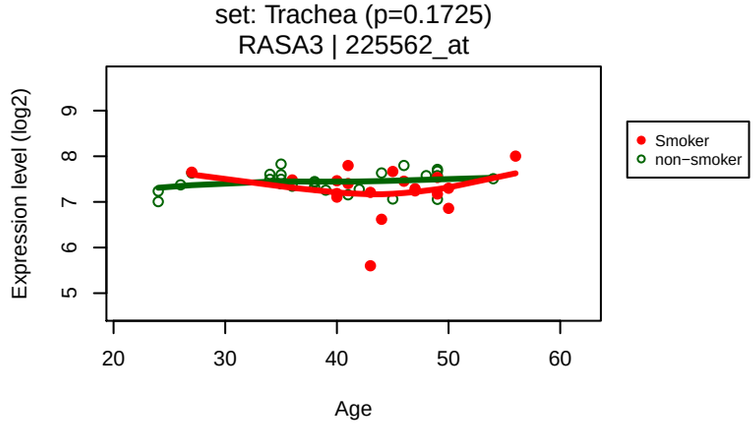

Gene expression ratio in Males
gene: RASA3 | 225562_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9042
p=0.1255
p=0.2472
p=0.7404
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RASA3 | 225562_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4939
p=0.1662
p=0.3137
p=0.6482
SmallAirway2
LargeAirway2
Trachea
Macrophage2

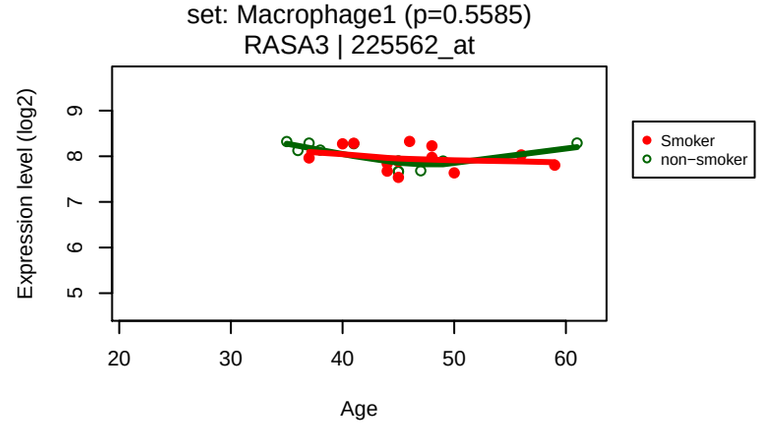

set: SmallAirway1 (p=0.8142)
RASA3 | 225562_at
Expression level (log2)
Age
Smoker
non−smoker

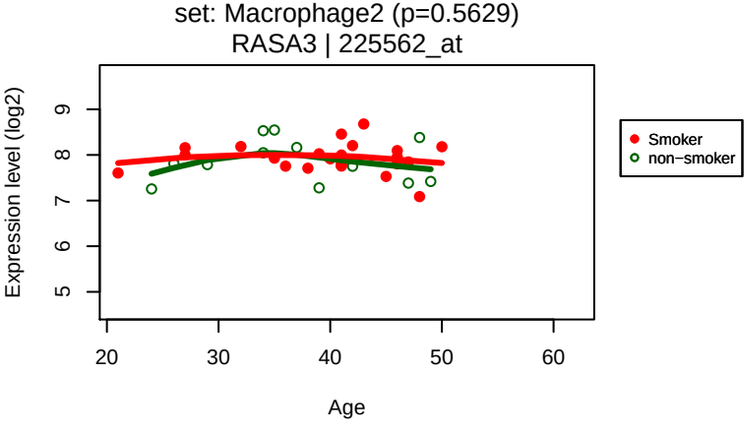

set: SmallAirway2 (p=0.9511)
RASA3 | 225562_at
Expression level (log2)
Age
Smoker
non−smoker

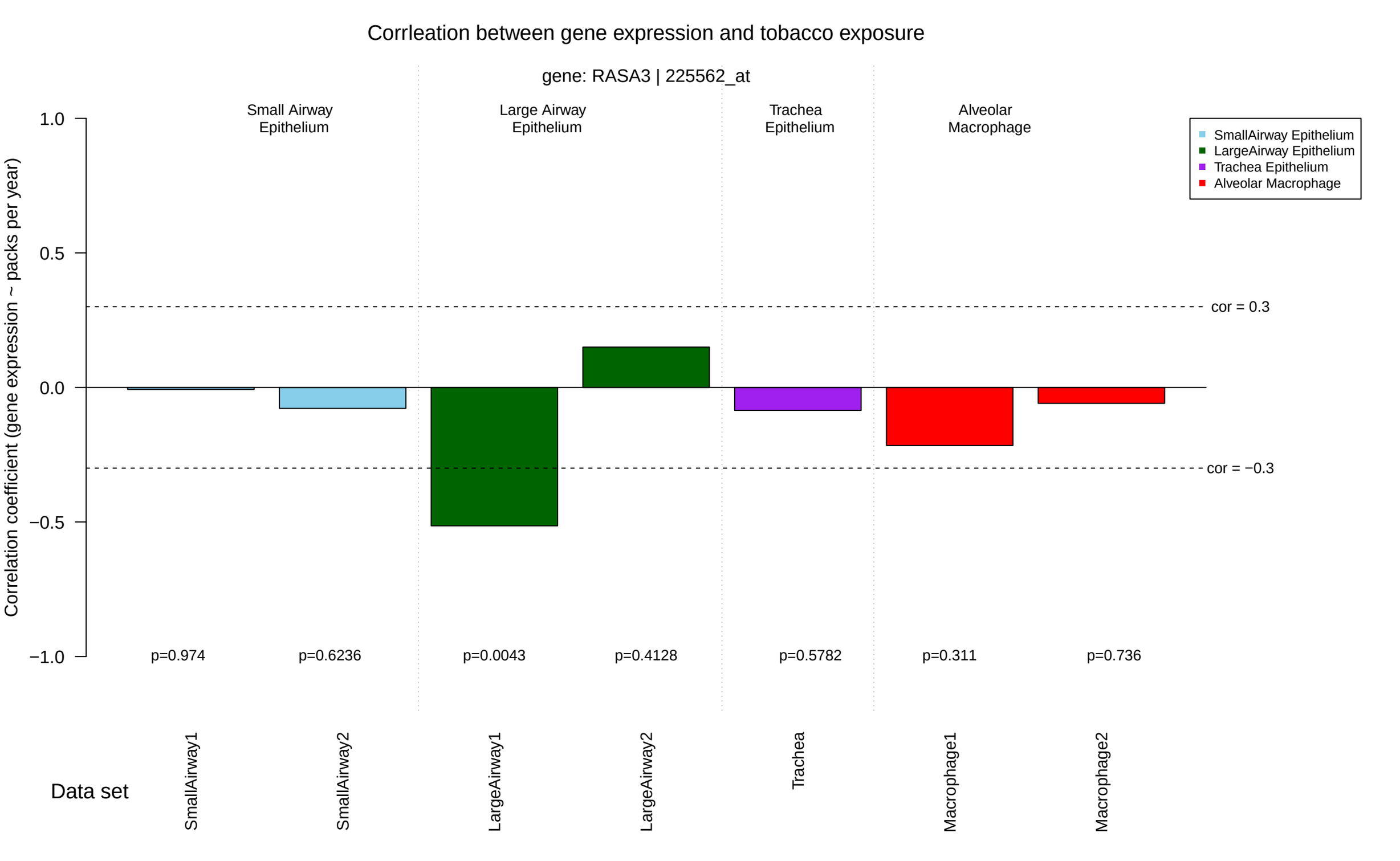

set: LargeAirway1 (p=0.0276)
RASA3 | 225562_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8202)
RASA3 | 225562_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1725)
RASA3 | 225562_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5585)
RASA3 | 225562_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5629)
RASA3 | 225562_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RASA3 | 225562_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.974
p=0.6236
p=0.0043
p=0.4128
p=0.5782
p=0.311
p=0.736
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2